T3 matched EFT Lagrangian terms: EFT_2_after_S

EFT order: 5; loop order: 1. Rows are model configurations and columns are field configurations. This stage integrates out S. Remaining active T3 fields: none.

Notation key

symbol	interaction / definition
g_1	$D_\mu \supset i g_1 Y B_\mu$
g_2	$D_\mu \supset i g_2 T^A W_\mu^A$
g_3	$D_\mu \supset i g_3 t^A G_\mu^A$
Y_u	$-\bar{Q} Y_u \widetilde{H} u + \text{h.c.}$
Y_d	$-\bar{Q} Y_d H d + \text{h.c.}$
Y_e	$-\bar{L} Y_e H e + \text{h.c.}$
y_1	$L F S_1 + \text{h.c.}$
y_2	$L F S_2 + \text{h.c.}$
M_F	$-\bar{F} M_F F$
T	$\mathrm{Tr}(Y_e^\dagger Y_e + 3 Y_u^\dagger Y_u + 3 Y_d^\dagger Y_d)$
$T_\nu^{(1)}$	$\mathrm{Tr}(y_1^\dagger y_1)$
$T_\nu^{(2)}$	$\mathrm{Tr}(y_2^\dagger y_2)$
T_ν	$T_\nu^{(1)} + T_\nu^{(2)}$
m_1^2	$m_1^2 S_1^\dagger S_1$
m_2^2	$m_2^2 S_2^\dagger S_2$
λ_1	$\frac{1}{2}\lambda_1 (H^\dagger H)^2$
$\lambda_{S_1}^{(1)}$	$\frac{1}{2}\lambda_{S_1}^{(1)} (S_1^\dagger S_1)^2$
$\lambda_{S_2}^{(1)}$	$\frac{1}{2}\lambda_{S_2}^{(1)} (S_2^\dagger S_2)^2$
$\lambda_{HS_1}^{(1)}$	$\lambda_{HS_1}^{(1)} (H^\dagger H)(S_1^\dagger S_1)$
$\lambda_{HS_2}^{(1)}$	$\lambda_{HS_2}^{(1)} (H^\dagger H)(S_2^\dagger S_2)$
$\lambda_{HS_1}^{(A)}$	$\lambda_{HS_1}^{(A)} (H^\dagger T^A H)(S_1^\dagger T^A S_1)$
$\lambda_{HS_2}^{(A)}$	$\lambda_{HS_2}^{(A)} (H^\dagger T^A H)(S_2^\dagger T^A S_2)$
λ_5	$\lambda_5 H H S_1 S_2^\dagger + \text{h.c.}$
$\lambda_{12}^{(1)}$	$\lambda_{12}^{(1)} (S_1^\dagger S_1)(S_2^\dagger S_2)$
$\lambda_{12}^{(A)}$	$\lambda_{12}^{(A)} (S_1^\dagger T^A S_1)(S_2^\dagger T^A S_2)$
$\lambda_{12}^{(\times)}$	crossed independent S_1–S_2 contraction
$\lambda_{S_1}^{(A)}$	additional independent S_1 self-contraction
$\lambda_{S_2}^{(A)}$	additional independent S_2 self-contraction
α	$Y_{S_1} = \alpha/2,\ Y_F = (\alpha+1)/2,\ Y_{S_2} = (\alpha+2)/2$
$C_2(d)$	$(d^2-1)/4$

$T(d)$	$d(d^2-1)/12$
$C^{ij;AB}_{LLS_1S_2}$	$L_iL_jS_1^AS_2^B$
C_{ijab}	$L_iL_j\phi_a\phi_b$
C_5^{ij}	$(L_i^TCL_j)HH$
R_W	$16\pi^2\beta_{C_5} \supset R_W\lambda_5 C_{LLS_1S_2}$

Field combinations 1 of 2

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	$H\,\ell$	$H\,\ell\,e$	$F\,H\,\ell$
Scotogenic-dS2-N	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	$+\frac{2\hbar\overline{y_{1,i_1,r_1}y_{2,i_2,r_1}}{L_{a_4,i_2}}^T C P_L L_{a_2,i_1} H_{a_1} H_{a_3} \lambda_5 \epsilon_{a_1,a_2}\epsilon_{a_3,a_4}}{\sqrt{3}\epsilon M_F}$ $+\frac{2\hbar\overline{y_{1,i_1,r_1}y_{2,i_2,r_1}}L_{a_4,i_2}{}^T C P_L L_{a_2,i_1} H_{a_1} H_{a_3}\lambda_5\epsilon_{a_1,a_2}\epsilon_{a_3,a_4}}{\sqrt{3}M_F}$ $+\frac{2\hbar\overline{\lambda_5 L_{a_3,i_1}}\, C P_R \overline{L_{a_4,i_2}}^T y_{1,i_1,r_1} y_{2,i_2,r_1}\epsilon_{a_1,a_3}\epsilon_{a_2,a_4} {H_{a_1}}^\dagger {H_{a_2}}^\dagger}{\sqrt{3}\epsilon M_F}$	$\frac{\hbar\overline{y_{1,i_3,r_1}}\,\overline{L_{a_1,i_2}}\, P_R\, e_{i_1} H_{a_1} y_{1,i_2,r_1} Y_{e,i_3,i_1}}{4\epsilon}$ $+\frac{3}{8}\hbar\overline{y_{1,i_3,r_1}}\,\overline{L_{a_1,i_2}}\, P_R\, e_{i_1} H_{a_1} y_{1,i_2,r_1} Y_{e,i_3,i_1}$ $+\frac{\hbar\overline{y_{2,i_3,r_1}}\,\overline{L_{a_1,i_2}}\, P_R\, e_{i_1} H_{a_1} y_{2,i_2,r_1} Y_{e,i_3,i_1}}{4\epsilon}$ $+\frac{3}{8}\hbar\overline{y_{2,i_3,r_1}}\,\overline{L_{a_1,i_2}}\, P_R\, e_{i_1} H_{a_1} y_{2,i_2,r_1} Y_{e,i_3,i_1}$	$+\frac{2\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{1,i_1,r_1}y_{2,i_2,r_1}}L_{a_4,i_2}{}^T C P_L L_{a_2,i_1} H_{a_1} H_{a_3}\lambda_5\epsilon_{a_1,a_2}\epsilon_{a_3,a_4}}{\sqrt{3}M_F}$ $+\frac{2\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{\lambda_5 L_{a_3,i_1}}\, C P_R \overline{L_{a_4,i_2}}^T y_{1,i_1,r_1} y_{2,i_2,r_1}\epsilon_{a_1,a_3}\epsilon_{a_2,a_4} {H_{a_1}}^\dagger {H_{a_2}}^\dagger}{\sqrt{3}M_F}$
cont.								$+\frac{2\hbar\overline{\lambda_5 L_{a_3,i_1}}\, C P_R \overline{L_{a_4,i_2}}^T y_{1,i_1,r_1} y_{2,i_2,r_1}\epsilon_{a_1,a_3}\epsilon_{a_2,a_4} {H_{a_1}}^\dagger {H_{a_2}}^\dagger}{\sqrt{3}M_F}$	$+\frac{\hbar\overline{y_{1,i_1,r_1}}\,\overline{Y_{e,i_3,i_2}}\,\overline{e_{i_2}}\, P_L\, L_{a_1,i_1} y_{1,i_3,r_1} {H_{a_1}}^\dagger}{4\epsilon}$ $+\frac{3}{8}\hbar\overline{y_{1,i_1,r_1}}\,\overline{Y_{e,i_3,i_2}}\,\overline{e_{i_2}}\, P_L\, L_{a_1,i_1} y_{1,i_3,r_1} {H_{a_1}}^\dagger$ $+\frac{\hbar\overline{y_{2,i_1,r_1}}\,\overline{Y_{e,i_3,i_2}}\,\overline{e_{i_2}}\, P_L\, L_{a_1,i_1} y_{2,i_3,r_1} {H_{a_1}}^\dagger}{4\epsilon}$	
cont.									$+\frac{3}{8}\hbar\overline{y_{2,i_1,r_1}}\,\overline{Y_{e,i_3,i_2}}\,\overline{e_{i_2}}\, P_L\, L_{a_1,i_1} y_{2,i_3,r_1} {H_{a_1}}^\dagger$	

Field combinations 2 of 2

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	F H ℓ e	
Scotogenic-dS2-N	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0		$+\frac{1}{4}\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{1,i_3,r_1}}\overline{L_{a_1,i_2}}\,P_R\,e_{i_1}H_{a_1}y_{1,i_2,r_1}Y_{e,i_3,i_1}$
									$+\frac{1}{4}\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{2,i_3,r_1}}\overline{L_{a_1,i_2}}\,P_R\,e_{i_1}H_{a_1}y_{2,i_2,r_1}Y_{e,i_3,i_1}$
									$+\frac{1}{4}\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{1,i_1,r_1}}\overline{Y_{e,i_3,i_2}}\,\overline{e_{i_2}}\,P_L\,L_{a_1,i_1}y_{1,i_3,r_1}{H_{a_1}}^\dagger$
								cont.	$+\frac{1}{4}\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{2,i_1,r_1}}\overline{Y_{e,i_3,i_2}}\,\overline{e_{i_2}}\,P_L\,L_{a_1,i_1}y_{2,i_3,r_1}{H_{a_1}}^\dagger$